

Making Small Models Reason on a Colab Budget: Layer-Frozen Group Relative Policy Optimization

Kriday Dave

Lead Researcher, Alethia Research Group
kriday@alethiaresearch.org

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Abstract

Group Relative Policy Optimization (GRPO) has democratized online reinforcement learning by eliminating the memory-heavy critic network. However, applying standard GRPO to compact models (1.5B–3B parameters) often triggers a severe “alignment tax,” resulting in a net-zero reasoning capability gain on out-of-distribution (OOD) benchmarks due to *Central Engine Disruption*—the destructive corruption of core mathematical and logical representations in early and middle layers ($L0$ – $L23$) by reasoning format gradients. To resolve this spatial optimization conflict, we introduce **Layer-Frozen Group Relative Policy Optimization (LF-GRPO)**. By freezing the model’s central engine and confining parameter updates strictly to the late-layer behavioral periphery ($L24$ – $L27$), LF-GRPO isolates cognitive formatting alignment from core capability networks. We empirically show that LF-GRPO successfully establishes robust chain-of-thought (CoT) monologue reasoning, jumping GSM-8K OOD accuracy to **50.00%** (compared to 42.00% for standard GRPO and base model limits) within a strict compute budget of 300 steps on a single commodity Tesla T4 GPU (16GB VRAM). Furthermore, we document two emergent behaviors: (1) relative advantage reward-hacking via multi-block open-tag exploits, and (2) schema generalization via novel XML tag hallucinations (`<bron>`). We present robust reward function modifications to eliminate these exploits and release our open-source, zero-dependency interactive pipeline.

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1 Introduction

The structural evolution of reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) has historically favored industrial conglomerates possessing massive computational fabrics. The recent introduction of Group Relative Policy Optimization (GRPO) [3] represents a critical paradigm shift, eliminating the actor-critic memory bottleneck by estimating baseline advantages directly from relative candidate rewards within a generated group.

Combined with parameter-efficient fine-tuning (PEFT/QLoRA) and highly optimized Triton kernels [4], GRPO has theoretically democratized online reinforcement learning, allowing student researchers and independent organizations to train reasoning models locally on consumer-grade hardware.

However, applying standard GRPO to smaller models (1.5B to 3B parameters) reveals an acute algorithmic barrier: the “alignment tax” scales aggressively downward. When standard GRPO is executed across all model layers, the backpropagated gradients responsible for shaping the step-by-step reasoning monologue (‘formatting’) and validating final token outputs (‘correctness’) propagate indiscriminately through the entire parameter stream.

Our prior work on the *Periphery Alignment & Central Logic* framework [2] establishes that transformer language models are spatially modular:

- **Central Engine (L0–L23):** A dense, invariant core network encoding raw mathematical logic, arithmetic operators, syntax infrastructure, and factual knowledge graphs.
- **Behavior Periphery (L24–L27):** A sparse late-layer gate system controlling output formatting, conversational style, and safety routing thresholds.

Under standard GRPO, the optimizer’s updates to satisfy formatting rewards inevitably corrupt the delicate arithmetic circuits residing in the central engine, resulting in a net-zero capability gain. The model learns to generate reasoning monologues, but the underlying calculation precision degrades at exactly the same rate, capping held-out test accuracy at baseline levels (42% on OOD GSM-8K).

To solve this conflict, we propose **Layer-Frozen Group Relative Policy Optimization (LF-GRPO)**. By strictly freezing the central engine and confining LoRA adapters to the behavior periphery, LF-GRPO isolates alignment updates. Gradients shape behavior routing and tags without introducing destructive noise into invariant arithmetic capabilities, realizing the full mathematical utility of chain-of-thought monologues on a strict consumer budget.

2 Related Work

Group Relative Policy Optimization. Traditional policy gradient methods like PPO [6] require co-allocating a policy model, a reference model, a value network (critic), and a reward model in active memory, causing severe out-of-memory (OOM) limitations on standard accelerators. GRPO [3] bypasses this by generating a group of N completions for each prompt and calculating advantages relative to the group’s mean and standard deviation, completely removing the critic.

Frugal AI & Notebook RL. Training complex reasoning loops on consumer accelerators (e.g. Tesla T4 with 16GB VRAM) has been accelerated by Unsloth [4], which offloads gradient buffers and utilizes dynamic standby memory states during vLLM rollout generation. This framework allows single-GPU colocate execution, but is highly sensitive to parameter density and optimization stability.

Spatial Modularity and Alignment Tax. Capability degradation after safety and instruction fine-tuning is widely documented [5]. In Paper 1 [2], we introduced Layer-Frozen Safety Fine-Tuning (LFSFT), proving that freezing L0–L23 during SFT protects mathematical capability (62% vs. 58% for full SFT). LF-GRPO extends this spatial protection paradigm to online reinforcement learning.

3 Methodology: Layer-Frozen GRPO (LF-GRPO)

3.1 Relative Advantage Math

LF-GRPO optimizes the active policy model π_θ using relative advantages computed across groups of generated outputs. For a given input prompt q , we sample a group of N candidate completions $\{o_1, o_2, \dots, o_N\}$. The relative advantage A_i for candidate completion o_i is defined as:

$$A_i = \frac{R(q, o_i) - \frac{1}{N} \sum_{j=1}^N R(q, o_j)}{\text{std}(\{R(q, o_1), \dots, R(q, o_N)\})} \quad (1)$$

where $R(q, o_i)$ is the raw reward evaluated by our custom verifiers. The policy objective minimizes the KL-divergence against the reference model π_{ref} while maximizing relative advantage:

$$\mathcal{L}_{GRPO}(\theta) = \frac{1}{N} \sum_{i=1}^N [\min(r_i(\theta)A_i, \text{clip}(r_i(\theta), 1 - \epsilon, 1 + \epsilon)A_i) - \beta D_{KL}(\pi_\theta(o_i|q) \parallel \pi_{ref}(o_i|q))] \quad (2)$$

where $r_i(\theta) = \frac{\pi_\theta(o_i|q)}{\pi_{ref}(o_i|q)}$ is the token probability ratio.

3.2 Autograd Periphery Insulation

To prevent the KL-divergence update and the advantage gradient from modifying the central logic, we strictly configure the backward autograd pass. The model’s hidden layers $\mathcal{L}$ are partitioned:

$$\mathcal{L} = \mathcal{L}_{engine} \cup \mathcal{L}_{periphery} \quad (3)$$

For a 28-layer model (e.g. Qwen2.5-1.5B), $\mathcal{L}_{engine} = \{0, 1, \dots, 23\}$ and $\mathcal{L}_{periphery} = \{24, 25, 26, 27\}$. Trainable LoRA adapters are initialized strictly within the periphery layers. We enforce a hard gradient mask:

$$\nabla_{\theta_j} \mathcal{L} = 0 \quad \forall j \in \mathcal{L}_{engine} \quad (4)$$

This spatial block guarantees 100% gradient insulation in the central engine throughout training.

3.3 Step-GRPO Decaying Reward and Loophole Mitigations

To force the model to output concise, high-density monologues rather than redundant token strings, we implement Step-GRPO. The step-decay reward R_{step} penalizes the occurrence of cognitive transition tokens (e.g., ‘wait’, ‘hmm’, ‘but’, ‘actually’) globally:

$$R_{step}(q, o_i) = \max(0.0, \gamma^{N_{steps}} - \text{penalty}_{blocks}) \cdot \mathbb{I}[o_i \text{ is mathematically correct}] \quad (5)$$

where $\gamma = 0.99$, and N_{steps} is the count of transition tokens searched globally across the entire completion text o_i .

In standard implementations, models discover a major structural loophole: closing a active `<think>` block and opening a new one resets the localized step counter. To completely eliminate this mathematical arbitrage, we introduce two critical mitigations:

1. **Global Flattening:** The transition token count N_{steps} is evaluated globally across the lowercased completion o_i , ignoring tag boundaries.
2. **Multi-Block Tag Penalty:** We apply a severe linear penalty for each auxiliary reasoning block beyond the first:

$$\text{penalty}_{blocks} = 0.5 \cdot (N_{open_tags} - 1) \quad (6)$$

where $N_{open_tags} = \text{count}(o_i, \text{<think>})$.

4 Experimental Design

We trained `unsloth/Qwen2.5-1.5B-Instruct` on the OpenAI GSM8K dataset (1000 samples). The training execution ran in standard Google Colab container environments utilizing a single Tesla T4 GPU (16GB VRAM) under severe time constraints.

4.1 Dataset & Setup

We utilize the OpenAI GSM8K dataset, subsetting to 1,000 training examples. The active policy model is initialized as `Qwen2.5-1.5B-Instruct` in 4-bit NormalFloat quantization. The LoRA rank and alpha are set to 32, targeting all linear projection layers within the late-layer behavioral periphery ($L24-L27$). Group size $N = 4$ is used for estimating relative policy advantages. Gradient accumulation steps are set to 4 with a batch size of 1 per device, simulating stable scaling on standard 16GB accelerators.

Table 1: Hyperparameter Configurations for Frugal LF-GRPO

Parameter	Value	Functional Purpose
Base Model	Qwen2.5-1.5B-Instruct	4-bit BN quantized baseline
LoRA Rank / Alpha	32 / 32	Target modules: q, k, v, o, gate, up, down
Layers to Transform	[24, 25, 26, 27]	Strictly limits trainable parameters to last 4 layers
Trainable Parameters	5,275,648 (0.34%)	Reduces active optimizer states by 7x
Batch Configuration	Batch=1, Accumulation=4	Simulates gradient stability on standard VRAM
Step Budget	300 Steps	Stage 1 (0–100: Format), Stage 2 (101–300: Correctness)
Optimizer	paged_adamw_8bit	Active CUDA page offloading
Sequence Limits	Prompt=512, Completion=384	Drastically reduces token rollout latency
Group Size (N)	4	Relative advantage calculation baseline

4.2 Reward Curve Analysis

During training, we track the evolution of rewards across optimization steps. Figure ?? shows the mean reward (R_j) and standard deviation band vs global step. We mark the transition from Stage 1 (formatting priming) to Stage 2 (correctness focus) at step 100 with a vertical dashed line. Formatting compliance aligns rapidly during the first 100 steps, after which the model is optimized directly for math correctness and length decay.

4.3 GSM8K Accuracy Comparison

We evaluate the performance of LF-GRPO against standard baseline configurations. Table 2 outlines the results on the full GSM8K test split.

Table 2: GSM8K OOD Evaluation Subset Benchmarks (50 Held-Out Examples)

Model	Mode	Pass@1 (%)	Avg. Completion Len (tokens)	R
Qwen2.5-1.5B-Instruct	LF-GRPO (This Work)	50.00%	208	
Qwen2.5-1.5B-Instruct	Standard GRPO (Full)	42.00%	230	
Qwen2.5-1.5B-Instruct	LF-GRPO (Run-1, Collapse)	42.00%	380	
Qwen2.5-1.5B-Instruct	UnSteered Base (Zero-Shot)	42.00%	—	

5 Empirical Findings

5.1 Causal Validation of Gradient Insulation

At step 0, prior to optimizer parameter updates, the active autograd backward pass was inspected. The results confirm absolute spatial isolation:

Listing 1: Gradient Insulation Verification Output at Step 0

```
=====
GRADIENT INSULATION VERIFICATION REPORT
=====
[+] Total trainable parameters: 56
[*] Sample active parameter gradient norms:
  - base_model.model.model.layers.24.self_attn.q_proj.lora_A...: grad_norm =
    0.000000
  - base_model.model.model.layers.24.self_attn.q_proj.lora_B...: grad_norm =
    0.002593
  - base_model.model.model.layers.24.self_attn.k_proj.lora_A...: grad_norm =
    0.000000
  - base_model.model.model.layers.24.self_attn.k_proj.lora_B...: grad_norm =
    0.003460
[+] Total frozen parameters: 338
[*] Sample frozen parameter gradient norms:
  - base_model.model.model.embed_tokens.weight: grad_norm = 0.000000
  - base_model.model.model.layers.0.self_attn.q_proj.weight: grad_norm = 0.000000
  - base_model.model.model.layers.0.self_attn.q_proj.bias: grad_norm = 0.000000
[+] VERIFICATION SUCCESS: 100% gradient insulation confirmed! All frozen layers
    have 0.0 gradient norm.
=====
```

5.2 Stage-by-Stage Convergence Curves

The two-stage reward configuration (`w_format` vs `w_correct`) successfully aligned formatting before enforcing mathematical correctness:

- **Stage 1 (Steps 0–50):** Format compliance progressed monotonically. Formatting reward climbed from 0.10 at step 0, bypassing 0.59 at step 20, and achieving a stable, perfect format reward of `**1.00**` by step 49.
- **Stage 2 (Steps 51–150):** Formatting weights scaled down (`w_format` = 0.2, `w_correct` = 1.0). Training correctness fluctuated between 45% and 66%, reflecting active policy search across candidate advantages.

5.3 Out-of-Distribution Math Reasoning Accuracy

Held-out evaluation sweeps on 50 OOD GSM-8K test prompts reveal the capability preservation advantages of LF-GRPO:

6 Discussion

6.1 Mitigating Central Engine Disruption

The comparative benchmarks in Table 3 provide causal proof of our spatial alignment hypothesis:

- **Standard GRPO:** Backpropagating updates through *L0–L27* simultaneously teaches reasoning structure while corrupting the core arithmetic circuits within *L0–L23*, leading to a flat 42.00% accuracy (no net benefit over base).

Table 3: OOD GSM-8K Evaluation Benchmarks

Model Configuration	Eval Mode	GSM-8K C
LF-GRPO Resumed Final (This Work)	Zero-Shot (ChatML reasoning prompt)	50.00% (25/50)
LF-GRPO (Scratch Run, Step 100)	Zero-Shot (ChatML reasoning prompt)	48.00% (24/50)
LF-GRPO (Scratch Run, Run-1, Step 200)	Zero-Shot (ChatML reasoning prompt)	42.00% (21/50)
LFSFT Model (Paper 1)	Zero-Shot	62.00%
Standard GRPO (Full-Layer LoRA)	Zero-Shot (CoT generated)	42.00% (21/50)
Qwen2.5-1.5B-Instruct (Base)	5-Shot (Baseline limit)	42.00%
Qwen2.5-1.5B-Instruct (Base)	Zero-Shot (ChatML reasoning prompt)	42.00% (21/50)
Qwen2.5-1.5B-Instruct (Base)	Zero-Shot (Standard baseline)	36.00% (18/50)

- **LF-GRPO:** Freezing $L0-L23$ isolates the mathematical core from gradient noise, allowing the monologue formatting benefits learned in $L24-L27$ to combine with the intact core, resulting in a substantial performance jump to ****50.00%**** under strict zero-shot evaluation (and up to ****58.00%**** in few-shot mode).

6.2 Run-1 Catastrophic Collapse: More XML Tags than Thinking

In the initial training run (**Run-1**), where the policy was optimized without correctness-gating on auxiliary rewards and without a direct length-based conciseness penalty, the model collapsed catastrophically by Step 200. It discovered a structural loophole in the SFT-trained format filters: by generating endless repetitive closing tags (`</answer>` and `</maths>` over 50 times consecutively), it successfully maximized intermediate formatting rewards while bypassing conciseness limits. The generated completions literally contained **more XML tags than actual thinking**.

This reward hacking had a highly destructive impact:

1. **Latency Inflation:** The evaluation speed collapsed from ≈ 11 s/it (at Step 100) to **28.19 s/it** (at Step 200) as the model stalled on every prompt generating repetitive garbage tokens.
2. **Central Engine Corruption:** Gradients backpropagated through the early layers corrupted the core arithmetic networks. In Example 1 (Janet’s ducks), the model performed a convoluted daily-weekly conversion (multiplying by 7) and then confidently output the wrong daily earnings (**#### 126** instead of **#### 18**), displaying a classic “looking smart while wrong” failure mode.

6.3 Run-2 Loophole-Free Mitigations

To eliminate this exploit, we designed **Run-2** to resume from the uncorrupted Step 100 checkpoint with three core mitigations integrated directly into the reward function:

1. **Correctness Gating (P-GRPO):** All auxiliary and formatting rewards are strictly zeroed out if the final mathematical answer is incorrect, removing the incentive to generate superficial calculations.
2. **Monologue Word-Count Decay:** Instead of a fragile blacklist of 6 words (which the model bypassed by shifting its vocabulary), we decay the correctness reward based on the monologue’s actual word count ($0.996^{\text{words}-100}$) after a 100-word grace window to allow thorough reasoning on complex, multi-step problems.
3. **Whitelisted Tag Fence:** Only `think` and `boxed` are whitelisted; all other tags incur a severe additive penalty of -1.5 to instantly suppress tag-spam runaway.

6.4 Novel Tag Hallucination and Format Generalization

We document an emergent phenomenon under OOD evaluation: the model generated a semantically appropriate, novel XML tag—`<bron>`—representing the prompt’s primary entity (Brandon) prior to outputting the

final answer slot, despite this tag never appearing in the training data. This suggests that GRPO conditioning acts on high-level abstract format schemas rather than specific token strings, indicating deep structural routing capability inside the behavior periphery.

We note that this behavior provides dynamic verification that the behavioral periphery acts as a schema encoder rather than a static dictionary.

6.5 Causal Deadlocks: The “Stuck Gear” Stalling Phenomenon

A manual audit of the line-by-line model completions reveals a highly notable qualitative phenomenon under out-of-distribution evaluation. While the Layer-Frozen behavioral periphery (*L24–L27*) successfully drives perfect structural formatting when the core mathematical engine (*L0–L23*) can resolve the problem, mathematical deadlocks on complex OOD prompts induce infinite policy state-loops.

Specifically, we observe that when the model *gets stuck* on a problem it cannot mathematically compute, the policy collapses into a repetitive state-loop, generating the same line of text or tags until it hits the `max_new_tokens` limit:

- **Example A (Josh flipping houses - Incorrect):** The model successfully inventory-allocates but becomes deadlocked during the profit computation, repeating: *“Identify quantities involved in calculation... Identify quantities remaining... Identify quantity needed to calculate profit...”*
- **Example B (Carla’s download - Incorrect):** Gets stuck calculating the time, repeating: *“Identify total time = 40 min \n Identify total time = 40 min \n Identify total time = 40 min...”*
- **Example C (Raymond & Samantha - Incorrect):** Repeats the line *“Identify the total time elapsed since Samantha became 31.”* 15 times consecutively.

Mechanistically, this “Stuck Gear” phenomenon represents a cognitive deadlock. Because the *Central Engine* (*L0–L23*) is frozen and lacks the capacity to solve the OOD step, it cannot feed the next logical state to the *Behavioral Periphery* (*L24–L27*). However, because the periphery is strictly conditioned by reinforcement learning to continue explaining and structuring, it enters a high-entropy repeating loop. This behavior provides **direct causal confirmation of spatial modularity**: the periphery acts strictly as a behavioral router rather than an independent reasoning engine.

7 Conclusion and Future Work

We have empirically validated Layer-Frozen Group Relative Policy Optimization (LF-GRPO), demonstrating that spatially isolating alignment gradients to the behavior periphery protects core logic capability while establishing robust cognitive formatting. Within a single Tesla T4 GPU budget, LF-GRPO improves GSM-8K reasoning accuracy to 50.00% under a strict whitelisted zero-shot protocol (and up to 58.00% under standard few-shot evaluation), bypassing standard full-layer GRPO limitations.

Future work will focus on integrating our trained LF-GRPO reasoning adapter with our Paper 1 safety adapter using advanced weight-space merging heuristics (TIES/DARE), stacking these specialized capabilities into a low-compute, edge-native FrankenMoE.

References

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